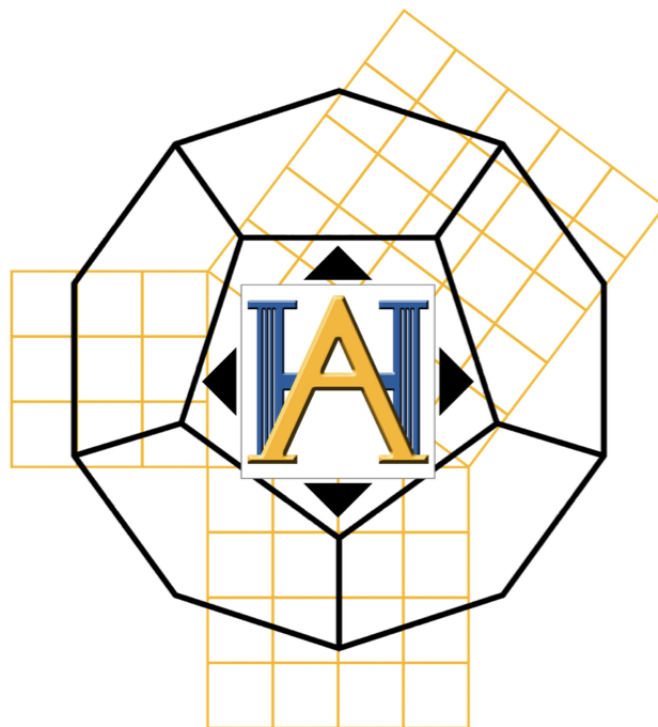




**AMERICAN HERITAGE SCHOOLS
COMPETITIVE MATHEMATICS**

2024 Mock Theta Individual #1

By Guilherme Salles and Rajam Arunprakash

1. What is $1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 8 + 10 + 11 + 12 + 14 + 15 + 16 + 17 + 18 + 19$?

(A) 170 (B) 176 (C) 189 (D) 190 (E) NOTA

2. Simplify the following, where $i = \sqrt{-1}$:

$$i + 2i^2 + 3i^3 + 4i^4 + 5i^5 + 6i^6 + 7i^7 + 8i^8$$

(A) $4 + 4i$ (B) $4 - 4i$ (C) $-4 - 4i$ (D) 0 (E) NOTA

3. Compute $\left| \begin{pmatrix} 4 & 15 \\ 0 & 6 \end{pmatrix} - \begin{pmatrix} 0 & 8 \\ -3 & 0 \end{pmatrix} \right|$.

(A) -3 (B) 0 (C) 3 (D) 48 (E) NOTA

4. Compute $(1 - i)^{21}$.

(A) -1024 (B) $-1024 + 1024i$ (C) $1024 - 1024i$ (D) $-1024i$ (E) NOTA

5. Find the sum of the digits of 4375_8 in base 2.

(A) 6 (B) 8 (C) 10 (D) 11 (E) NOTA

6. Which of the following best describes the locus of complex numbers z such that $||z - 3 + 4i| - |z + \ln 2|| = \pi^e$?

(A) Circle (B) Non-circular Ellipse (C) Parabola (D) Hyperbola (E) NOTA

7. Let $S_n = An^2 + Bn + C$ for real A, B, C and $n \geq 1$. If $S_1 = 4$, $S_2 = 10$, and $S_3 = 22$, find $A - B + C$.

(A) 4 (B) 10 (C) 13 (D) 14 (E) NOTA

8. An “unknown” NSU professor wrote a 20 question Theta Individual with 20% Geometry questions, 80% Algebra questions. When editing the test, Mr. Ariew adds A Algebra questions and C Combo questions. If the test is now 10% Geometry and 60% Algebra, then what is (A, C) ?

(A) (2, 8) (B) (4, 16) (C) (8, 12) (D) (24, 16) (E) NOTA

9. What is the remainder when the function $x^{2024} - 81x^{2022} + x^2 - 6x + 5$ is divided by $x - 9$?

(A) 0 (B) 32 (C) 81 (D) 140 (E) NOTA

10. A set of Tarot cards consists of 22 distinct *Major Arcana*, or cards. Each card has its own attributes and interpretations; however, when the card is upside-down, or “reversed”, its meaning changes. A “Three-Card Spread” involves having 3 distinct, each of which can be reversed, Tarot cards representing one’s past, present, and future. If the prime factorization for the number of unique Three-Card Spreads, given that all 3 cards are *Major Arcana*, is equal to $2^a \times 3^b \times 5^c \times 7^d \times 11^e$, find $a + b + c + d + e$.

(A) 7 (B) 8 (C) 9 (D) 10 (E) NOTA

11. Find the sum of the coefficients of $(2w + 2x - 2y - 3z)^5$ that have an x term.

(A) -1 (B) 32 (C) 242 (D) 244 (E) NOTA

12. Let a , b , and c be the roots of the polynomial $x^3 + x^2 + 12x - 3$. Find

$$\frac{b+c}{a^2+a} + \frac{a+c}{b^2+b} + \frac{a+b}{c^2+c}$$

(A) -4 (B) -2 (C) 2 (D) 4 (E) NOTA

13. Circle ω , which is outside $\triangle RAJ$, is tangent to $\overline{AJ}$. Sides $\overline{RA}$ and $\overline{RJ}$ are extended and are tangent to circle ω at points A' and J' respectively. If $RA' = 32$ and $RJ' = 40$, what is the perimeter of $\triangle RAJ$?

(A) 64 (B) 72 (C) 80 (D) 108 (E) NOTA

14. Given the following system of equations, find x :

$$w + 4x - 6y + z = 4$$

$$-2w + 7y - 9z = 0$$

$$3w + 9x - y + 5z = 9$$

$$-8w - x + y - z = -1$$

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 3 (E) NOTA

15. Find the number of integer values of $\frac{y}{x}$ such that

$$\frac{1}{512} < \frac{x}{x+y} < \frac{5}{2048}$$

(A) 100 (B) 101 (C) 102 (D) 103 (E) NOTA

16. Consider $\triangle BLS$ with strictly integer side lengths. If $\angle B \approx 10^\circ$ and $LS = 1$, which of the following is closest to the measure of $\angle S$?

(A) 84° (B) 85° (C) 86° (D) 88° (E) NOTA

Use the following information for the rest of the questions on this test: Frazer is my favorite test writer! His questions are always so fun and creative and always feel so original! Let's celebrate Frazer's amazing test writing by doing some of his questions.

17. (2021 Alpha Individual Nationals #23) If $2^{10} + 2^5 = K^5 + K^{10}$, what is the sum of all possible values of K^5 ?

(A) -1 (B) 1 (C) 32 (D) 33 (E) NOTA

18. (2021 Alpha Individual Nationals #4) In triangle XYZ , angle Y has a tangent of $\frac{1}{5}$. $XY = 2\sqrt{26}$ and $YZ = 8$. Find the length of XZ .
- (A) $2\sqrt{34}$ (B) $8\sqrt{2}$ (C) $2\sqrt{2}$ (D) 2 (E) NOTA
19. (2021 Alpha Individual Nationals #10) Diagonal RU of rectangle $MRLU$ is divided into three segments of length 1 by parallel lines that pass-through M and L and are perpendicular to RU . The area of rectangle $MRLU$ is?
- (A) 4.5 (B) $1 + 2\sqrt{3}$ (C) $2\sqrt{3}$ (D) $3\sqrt{2}$ (E) NOTA
20. (2023 Alpha Cipherring Nationals #3) Find the product of the real roots of the equation

$$x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$$

- (A) 15 (B) 20 (C) 36 (D) 720 (E) NOTA